

Alexey Grekov

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Education

Bachelor of Science in Software Engineering

Expected May 2027

Schulich School of Engineering | University of Calgary

- **Certificates:** Full-Stack Engineer Career Path — Codecademy, Engineering Leadership Program Tier 3 — University of Calgary

Experience

Undergraduate Thesis Researcher - Power Systems

September 2026 - Present

PSCAD Simulation Automation for Substation Reliability | University of Calgary

- Writing an undergraduate thesis under Dr. Xi Wang on substation grid reliability and condition monitoring, automating PSCAD simulation testing through scripted case generation to replace manual GUI setup

Capstone Project – Undergraduate Researcher

September 2026 - Present

Agentic Simulator for Emergency Department Operations | University of Calgary

Project Overview

- Completing software engineering capstone project under the supervision of Dr. Steve Drew, contributing to the development of an LLM-powered emergency department simulator

Electrical Engineering Intern

September 2025 - August 2026

ATCO Electric Yukon | Whitehorse, Yukon

- Designed overhead and underground distribution grids valued up to \$6M by evaluating voltage drop, amperage, and transformer loading constraints. Authored capital project proposals comparing construction costs against electrical performance to justify the optimal design
- Mapped voltage drop across the city of Whitehorse to determine breakpoints for every single distribution substation and downstream feeder. Gathered equipment Vpu data using CYME and automated calculations on the exported CSVs using Power Query in Excel
- Conducted a Yukon wide coordination study analyzing circuit breaker and recloser settings against simulated load flow data in CYME to calculate equipment loading ratios. Identified 10 devices requiring winter setting adjustments to prevent non fault trips
- Modernized engineering, construction, and billing workflows by writing JavaScript for Adobe forms and updating legacy Excel VBA scripts

Server Infrastructure Intern

May 2025 - August 2025

Alberta Health Services | Calgary, Alberta

- Developed Grafana dashboards using LogQL to visualize critical server metrics such as kernel high-severity logs, failed SSH logins, and out-of-memory events, streamlining triage and root cause analysis during weekly team meetings
- Engineered an automated Python ETL pipeline querying PostgreSQL for unpatched servers, serving the processed data via a Podman-containerized Nginx web server to live Grafana dashboards using a monthly cron job
- Authored a Podman adoption guide for development teams, detailing migration strategies from Docker to Podman and Docker Compose conversion using the Kompose tool
- Created clear documentation in Microsoft Visio outlining patching workflows for Unix (IBM AIX) and Linux (RHEL and OEL) servers

Technical Team

Tech Start UCalgary

September 2025 - January 2026

Front End Developer | University of Calgary

- Migrated a legacy client side React website to a modern Next.js architecture, utilizing TypeScript and Node.js for server side handling
- Created Figma prototypes for a newly integrated dashboard feature, designing pages for signup, user profiles, and announcements

Projects

Scribble Squad Pictionary

February 2026 - March 2026

TypeScript, Tailwind, React, Redux, Vite, Express, Node.js, Railway

- Built a real time multiplayer Pictionary style web game deployed on Railway, utilizing WebSockets to broadcast game state snapshots that synchronize live canvas strokes, chat messages, and player scoring across clients without polling
- Enhanced the user experience with animations powered by Vanta.js and Three.js, while integrating the Web Audio API for synthesized game sound effects
- Deployed independent frontend and backend services to Railway, implementing fault tolerance logic to handle mid game player disconnections and maintain uninterrupted gameplay